AI/ML Engineer | Generative AI Specialist | Global Production Technical Lead

PROFESSIONAL PROFILE

Results-driven AI/ML Engineer with 10+ years of experience at HSBC, combining deep expertise in enterprise technology with a growing specialisation in Artificial Intelligence, Machine Learning, and Generative AI. Proven track record in designing and deploying AI-powered solutions, building predictive models, developing chatbots and intelligent agents, and implementing RAG-based applications leveraging institutional knowledge sources. Hands-on experience with Python, LLMs, Vertex AI, Prompt Engineering, and end-to-end ML pipelines. Adept at translating complex business challenges into scalable AI solutions while maintaining strong stakeholder engagement, governance, and cross-functional collaboration across global teams.

CORE TECHNICAL SKILLS

AI / ML	Generative AI & LLMs	Cloud & MLOps
Python (NumPy, Pandas, Sklearn)	OpenAI / Gemini / Claude APIs	Google Vertex AI
Machine Learning Pipelines	Prompt Engineering	Google Cloud Platform (GCP)
Predictive Modelling	RAG Architectures	Model Deployment & Monitoring
NLP & Text Analytics	LangChain / LlamaIndex	CI/CD for ML
AI Agent Development	Chatbot & Virtual Assistants	API Integrations

ADDITIONAL COMPETENCIES

Project & Programme Management	Incident & Problem Management	Team Leadership
Agile / SAFe / Waterfall	ITIL Process Management	Budget & Financial Forecasting
RAID Management	Stakeholder Engagement	Data Analysis & Trend Identification
Site Reliability Engineering	Root Cause Analysis	Knowledge Management

CAREER SUMMARY

Key Responsibilities – AI/ML Engineering

- Design, develop, and deploy AI and machine learning solutions to automate production support workflows, reducing manual effort and resolution times across HSBC's global technology estate.
- Build and tune predictive models (classification, regression, anomaly detection) and Generative AI applications to surface insights from enterprise operational data.
- Develop AI-powered assistants, intelligent chatbots, and autonomous agents that handle Tier-1 incident triage, FAQ resolution, and knowledge retrieval for internal teams.
- Implement RAG (Retrieval-Augmented Generation) based solutions leveraging HSBC's institutional knowledge bases, policy documents, and ITIL process repositories to power context-aware AI responses.
- Deploy and manage AI applications on Google Vertex AI, configuring pipelines, endpoints, and monitoring dashboards to ensure model performance in production.
- Monitor and optimise AI models in production environments; establish drift detection, logging, and retraining schedules to maintain accuracy and reliability at scale.

- Integrate AI APIs (OpenAI, Gemini, Vertex AI) with internal platforms via REST/gRPC APIs, enabling seamless embedding of AI capabilities into existing enterprise tooling.
- Apply Prompt Engineering techniques to design effective system prompts and few-shot examples, improving LLM task accuracy for summarisation, classification, and code generation use cases.

Key Responsibilities – Programme Management & Leadership

- Lead end-to-end delivery and governance of large-scale technology projects, coordinating across development, testing, and deployment teams globally.
- Manage RAID (Risks, Assumptions, Issues, Dependencies) logs and maintain comprehensive project plans aligned to milestones and executive reporting cadences.
- Drive major incident management and permanent fix initiatives through strategic automation, reducing recurring production issues by 30%.
- Manage project budgets including Capex/Opex forecasting, supplier contracts, and resource utilisation from onboarding to go-live.
- Facilitate cross-functional knowledge sharing, training sessions, and documentation to maximise team capability and tool adoption.

KEY AI/ML PROJECTS AT HSBC

Intelligent Incident Triage Chatbot *2024 – Present*

- Built a Generative AI chatbot using Python, LangChain, and Vertex AI to automate Tier-1 incident classification and suggested resolution routing, reducing triage time by 40%.
- Implemented RAG pipeline ingesting HSBC's internal runbooks, knowledge base articles, and ITIL procedures, enabling the bot to surface contextually accurate resolution steps.
- Deployed on Google Cloud via Vertex AI endpoints with monitoring dashboards; integrated with ServiceNow via REST APIs for ticket auto-creation and enrichment.

Production Anomaly Detection – Predictive Model *2023 – 2024*

- Designed and trained an ML model (Python / Scikit-learn / XGBoost) on historical incident telemetry data to predict production outages 30–60 minutes in advance.
- Integrated model outputs into a real-time alerting pipeline, enabling proactive incident response and reducing Mean Time to Detect (MTTD) by 35%.
- Managed model lifecycle on Google Vertex AI including versioning, A/B testing of model variants, and scheduled retraining against fresh telemetry.

Knowledge Base Q&A System (RAG) *2023*

- Built an internal Q&A assistant using RAG architecture (LangChain + Vertex AI Embeddings + Vector Search) over 5,000+ HSBC policy and process documents.
- Enabled teams to query institutional knowledge in natural language, reducing knowledge-retrieval time by 60% and decreasing inbound support queries by 25%.
- Applied prompt engineering to tune LLM responses for factual grounding and appropriate citation of source documents.

Automated Root Cause Analysis (AI-Powered RCA) *2022 – 2023*

- Developed an NLP-based tool using Python and OpenAI APIs to analyse incident logs, identify patterns, and draft RCA summaries, cutting manual RCA effort by 50%.
- Integrated output into HSBC's problem management workflow, with human-in-the-loop review before finalisation, maintaining governance and compliance standards.

KEY ACHIEVEMENTS

- Reduced incident triage time by 40% through deployment of AI-powered chatbot with RAG-backed knowledge retrieval.

- Cut production outage detection time by 35% via ML-based anomaly prediction integrated into live monitoring pipelines.
- Achieved 25% improvement in project delivery timelines by introducing lean governance and automation-first methodologies.
- Decreased incident resolution times by 30% through collaboration, process automation, and AI-assisted diagnostics.
- Reduced knowledge-retrieval time by 60% via internal Q&A system built on RAG architecture over institutional documents.
- Lowered stakeholder-reported incident volume by 65% through proactive best-practice sharing and AI-driven monitoring.

EDUCATION

Bachelor of Engineering – Green Fort Engineering College, Jawaharlal Nehru Technological University

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Google Cloud Professional Machine Learning Engineer (in progress)
- Generative AI on Vertex AI – Google Cloud Skills Boost
- Certification in Leadership Skills for a Manager – Saudi Human Resources
- Internal Auditor Participation Certification – Intertek Mody